

Vector spaces

Vectors are things we can add and scale.

Examples.

- The set of $m \times n$ matrices: if $A, B \in M_{m \times n}$, then $2A + B \in M_{m \times n}$.
- Polynomials. If

$$f(x) = a_0 + a_1x + \cdots + a_nx^n, \quad g(x) = b_0 + b_1x + \cdots + b_mx^m, \quad m < n,$$

then

$$(f + g)(x) = a_0 + b_0 + (a_1 + b_1)x + \cdots + (a_m + b_m)x^m + a_{m+1}x^{m+1} + \cdots + a_nx^n,$$

and

$$2f(x) = 2a_0 + \cdots + 2a_nx^n.$$

- Sequences of numbers (x_n) , where n is an integer. For example,

$$2^n = \{\dots, 2^{-1}, 2^0, 2^1, 2^2, \dots\},$$

and addition and scalar multiplication are defined term-by-term:

$$(2x + y)_n = 2x_n + y_n.$$

- Functions $f : X \rightarrow \mathbb{R}$, with

$$(f + g)(x) = f(x) + g(x), \quad (2f)(x) = 2f(x).$$

Definition. Let V be a set whose elements are called vectors, with two operations:

1. $V \times V \rightarrow V$, denoted $(v, w) \mapsto v + w$ (**addition**);
2. $\mathbb{R} \times V \rightarrow V$, denoted $(c, v) \mapsto cv$ (**scaling**).

For all $u, v, w \in V$ and $c, d \in \mathbb{R}$, the following vector-space properties hold:

1. $u + v = v + u$.
2. $(u + v) + w = u + (v + w)$.
3. There is a zero vector 0 with $v + 0 = v$ for every $v \in V$.
4. For every $v \in V$, there is $-v$ with $v + (-v) = 0$.
5. $c(u + v) = cu + cv$.
6. $(c + d)u = cu + du$.
7. $(cd)u = c(du)$.
8. $1v = v$.

Useful consequences:

9. The zero vector is unique.
10. v uniquely determines $-v$.
11. $0v = 0$.
12. $c0 = 0$.
13. $-v = (-1)v$.

Proof of (9). If $0'$ is another zero vector, then

$$0 = 0 + 0' = 0'.$$

Proof of (10). If $-v$ and v' both satisfy $-v + v = 0 = v' + v$, then

$$-v = -v + 0 = -v + (v + v') = 0 + v' = v'.$$

Example. Any subspace of $\mathbb{R}^n$ is a vector space.

Note. In the definition of a vector space, the scalars need not come from $\mathbb{R}$. One can use another field, such as $\mathbb{C}$ or $\mathbb{Q}$.

There is also a related notion of a **module**, where scalars can come from objects such as $\mathbb{Z}$ or $\mathbb{Z}/n\mathbb{Z}$ when n is not prime, or from polynomials.

Topics in vector spaces

1. Linear combinations and subspaces.
2. Linear independence and span.
3. Linear maps, the image of T , and the kernel of T .
4. Bases, coordinates, and dimension.

Definition. Let S be a set of vectors in a vector space V . The **span** of S is the set of all linear combinations of vectors in S .

The set S need not be finite. A linear combination has the form

$$c_1 v_1 + \cdots + c_k v_k.$$

Example.

$$S = \{1, x^2, x^4, \dots\}.$$

Then $\text{span } S$ is the set of polynomials with no odd monomials.

Definition. A **basis** of a vector space V is a possibly infinite set β of vectors in V such that

$$V = \text{span } \beta$$

and, for every finite selection $v_1, \dots, v_k \in \beta$,

$$c_1 v_1 + \cdots + c_k v_k = 0 \implies \begin{bmatrix} c_1 \\ \vdots \\ c_k \end{bmatrix} = \begin{bmatrix} 0 \\ \vdots \\ 0 \end{bmatrix}.$$

Example. A basis for the vector space of all polynomials with real coefficients is

$$\beta = \{1, x, x^2, \dots, x^n, \dots\}.$$

Theorem. The set $\{1, x, x^2, \dots, x^n, \dots\}$ is independent.

Proof. Suppose

$$c_1 x^{i_1} + \cdots + c_k x^{i_k} = 0, \quad i_1 < i_2 < \cdots < i_k.$$

Divide by x^{i_1} to obtain

$$c_1 + c_2 x^{i_2 - i_1} + \cdots + c_k x^{i_k - i_1} = 0.$$

Setting $x = 0$ gives $c_1 = 0$; induction on k finishes the proof.

Corollary. The set

$$\{1, x, x^2, \dots, x^n, \dots\}$$

is a basis of $V = \mathbb{R}[x]$, the vector space of polynomials with real coefficients.

Definition. A **subspace** of a vector space is a subset S such that

$$x, y \in S \implies x + y \in S, \quad c \in \mathbb{R}, x \in S \implies cx \in S.$$

Note. A subspace of a vector space is itself a vector space.

Question. Do polynomials of degree exactly n form a subspace?

No. If

$$a_0 + a_1x + \dots + a_nx^n$$

has degree n , then subtracting a_nx^n gives

$$a_0 + a_1x + \dots + a_{n-1}x^{n-1},$$

which does not have degree n .

Question. What is a basis of $\mathbb{R}_{\leq n}[x]$?

$$\beta = \{1, x, x^2, \dots, x^n\}.$$

Example. Do upper triangular $n \times n$ matrices form a subspace of the vector space of all $n \times n$ matrices?

Yes. For example,

$$\begin{bmatrix} 1 & 2 \\ 0 & 3 \end{bmatrix} + 5 \begin{bmatrix} -1 & 1 \\ 0 & 2 \end{bmatrix}$$

is upper triangular; addition and scalar multiplication preserve the zero entries below the diagonal.

Example. A basis of $M_{m \times n}(\mathbb{R})$, the vector space of $m \times n$ matrices, consists of the matrices having exactly one nonzero entry, equal to 1.

For instance,

$$\begin{bmatrix} 2 & 1 & 3 \\ 5 & 1 & 7 \end{bmatrix} = 2 \begin{bmatrix} 1 & 0 & 0 \\ 0 & 0 & 0 \end{bmatrix} + \begin{bmatrix} 0 & 1 & 0 \\ 0 & 0 & 0 \end{bmatrix} + \cdots.$$

Therefore,

$$\dim M_{m \times n}(\mathbb{R}) = mn.$$

Definition. The **dimension** of a vector space is the number of vectors in a basis.

Examples.

$$\dim \mathbb{R}_{\leq n}[x] = n + 1,$$

because its elements have the form $a_0 + a_1x + \cdots + a_nx^n$.

$$\dim \mathbb{R}[x] = \infty,$$

because $\{1, x, x^2, \dots, x^n, \dots\}$ is a basis.

Examples.

$$\dim M_{n \times n}(\mathbb{R}) = n^2.$$

The vector space of $n \times n$ upper triangular matrices has dimension

$$n + \frac{n^2 - n}{2} = \frac{n^2 + n}{2}.$$

Definition. A map $T : V \rightarrow W$, where V, W are vector spaces, is **linear** if

$$T(c_1 v_1 + c_2 v_2) = c_1 T(v_1) + c_2 T(v_2)$$

for every $v_1, v_2 \in V$ and $c_1, c_2 \in \mathbb{R}$.

Example. Define

$$T : M_{n \times n}(\mathbb{R}) \rightarrow M_{n \times n}(\mathbb{R}), \quad T(A) = A^T.$$

Then

$$T(c_1 A_1 + c_2 A_2) = (c_1 A_1 + c_2 A_2)^T = c_1 A_1^T + c_2 A_2^T = c_1 T(A_1) + c_2 T(A_2),$$

so T is linear.

Example. Define

$$T : \mathbb{R}[x] \rightarrow \mathbb{R}[x], \quad T(f) = f'.$$

Then

$$T(c_1f_1 + c_2f_2) = (c_1f_1 + c_2f_2)' = c_1f_1' + c_2f_2' = c_1T(f_1) + c_2T(f_2).$$

Thus T is linear.

Example. For this derivative map,

$$\ker T = \{f \in \mathbb{R}[x] : f' = 0 = T(f)\},$$

which is the set of constant polynomials.

Example. Define

$$T : \mathbb{R}[x] \rightarrow \mathbb{R}[x], \quad T(f) = \int_c^x f(t) dt.$$

Since

$$\int_c^x (c_1f_1 + c_2f_2) dt = c_1 \int_c^x f_1 dt + c_2 \int_c^x f_2 dt,$$

T is linear.

Definition. A vector space with a finite basis is **finite-dimensional**.

Theorem. Let

$$\beta = \{v_1, \dots, v_n\}$$

be a basis of a vector space V . Let $T : V \rightarrow \mathbb{R}^n$ be the coordinate map

$$T(x) = [x]_\beta.$$

That is,

$$T(c_1v_1 + \dots + c_nv_n) = \begin{bmatrix} c_1 \\ \vdots \\ c_n \end{bmatrix}.$$

Then T is linear and invertible.

Definition. A linear invertible map is a **linear isomorphism**.

If $T : V \rightarrow W$ is a linear isomorphism, then V and W are **isomorphic**.

Example. The map

$$a_0 + a_1x + \dots + a_nx^n \mapsto \begin{bmatrix} a_0 \\ \vdots \\ a_n \end{bmatrix}$$

is a linear isomorphism from $\mathbb{R}_{\leq n}[x]$ to $\mathbb{R}^{n+1}$. The basis is

$$\beta = \{1, x, \dots, x^n\}.$$

Key point. In isomorphic vector spaces, linear algebra works in the same way.

Theorem. If $T : V \rightarrow W$ is linear and invertible, then

$$T^{-1} : W \rightarrow V$$

is linear.

Proof. Suppose, for a contradiction, that

$$T^{-1}(c_1w_1 + c_2w_2) \neq c_1T^{-1}(w_1) + c_2T^{-1}(w_2).$$

Since T is one-to-one, applying T preserves the inequality. But

$$\begin{aligned} T(T^{-1}(c_1w_1 + c_2w_2)) &= c_1w_1 + c_2w_2 \\ &= c_1T(T^{-1}(w_1)) + c_2T(T^{-1}(w_2)) \\ &= T(c_1T^{-1}(w_1) + c_2T^{-1}(w_2)), \end{aligned}$$

which is a contradiction.

Definition. Let S be the space of sequences $(x_k)_{k \geq 0}$ with $x_k \in \mathbb{R}$. Define

$$\text{Fib} = \{(x_k) \in S : x_{k+2} = x_{k+1} + x_k\}.$$

For example, if $(x_k) \in \text{Fib}$, then

$$2(x_k) = 2(x_k), \quad 2x_{k+2} = 2x_{k+1} + 2x_k,$$

so scaling preserves the recurrence. Likewise, addition preserves the recurrence. Thus Fib is a subspace of S .

Define

$$T : \text{Fib} \rightarrow \mathbb{R}^2, \quad (x_k) \mapsto \begin{bmatrix} x_0 \\ x_1 \end{bmatrix}.$$

This map is linear and invertible: the recurrence determines every later term, for example,

$$x_2 = x_0 + x_1, \quad x_3 = x_1 + x_2.$$

Therefore Fib is linearly isomorphic to $\mathbb{R}^2$, and a basis can be obtained from

$$\{T^{-1}(e_1), T^{-1}(e_2)\}.$$

To find vectors in Fib, search for sequences of the form r^k , where $r \neq 0$. The recurrence gives

$$r^{k+2} = r^{k+1} + r^k,$$

so

$$r^2 - r - 1 = 0, \quad r_{\pm} = \frac{1 \pm \sqrt{5}}{2}.$$

Thus

$$(r_+^k), \quad (r_-^k)$$

belong to Fib and are independent, so they form a basis of Fib.

Every Fibonacci sequence has the form

$$f_k = C_+ \left(\frac{1 + \sqrt{5}}{2} \right)^k + C_- \left(\frac{1 - \sqrt{5}}{2} \right)^k.$$

For the sequence with $f_0 = 0$ and $f_1 = 1$,

$$0 = C_+ + C_-, \quad 1 = C_+ r_+ + C_- r_-.$$